

Counting Bits

Time limit: 1.00 s

Memory limit: 512 MB

Your task is to count the number of one bits in the binary representations of integers between 1 and n .

Input

The only input line has an integer n .

Output

Print the number of one bits in the binary representations of integers between 1 and n .

Constraints

- $1 \leq n \leq 10^{15}$

Example

Input	Output
7	12

Explanation: The binary representations of 1 ... 7 are 1, 10, 11, 100, 101, 110, and 111, so there are a total of 12 one bits.